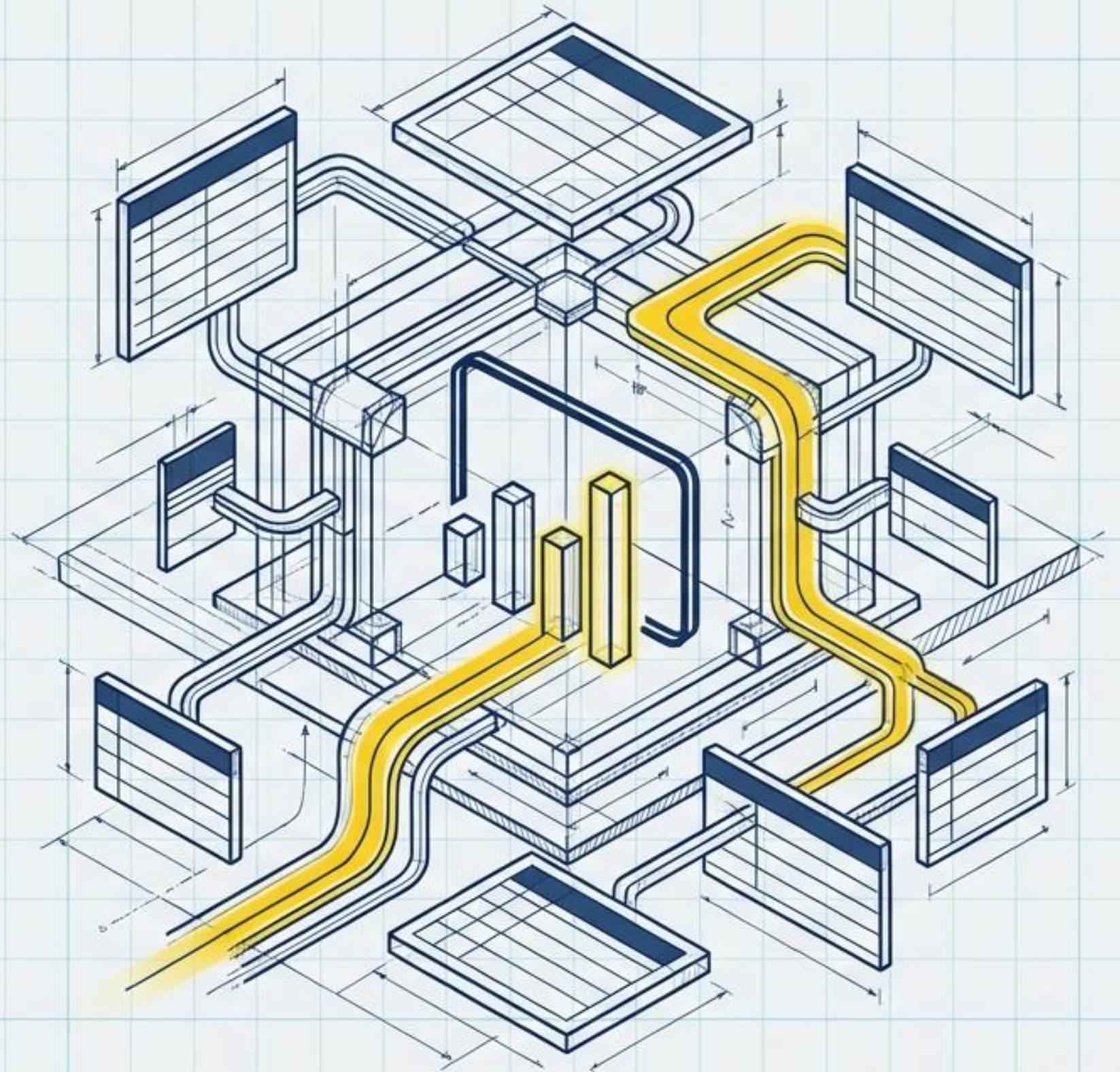


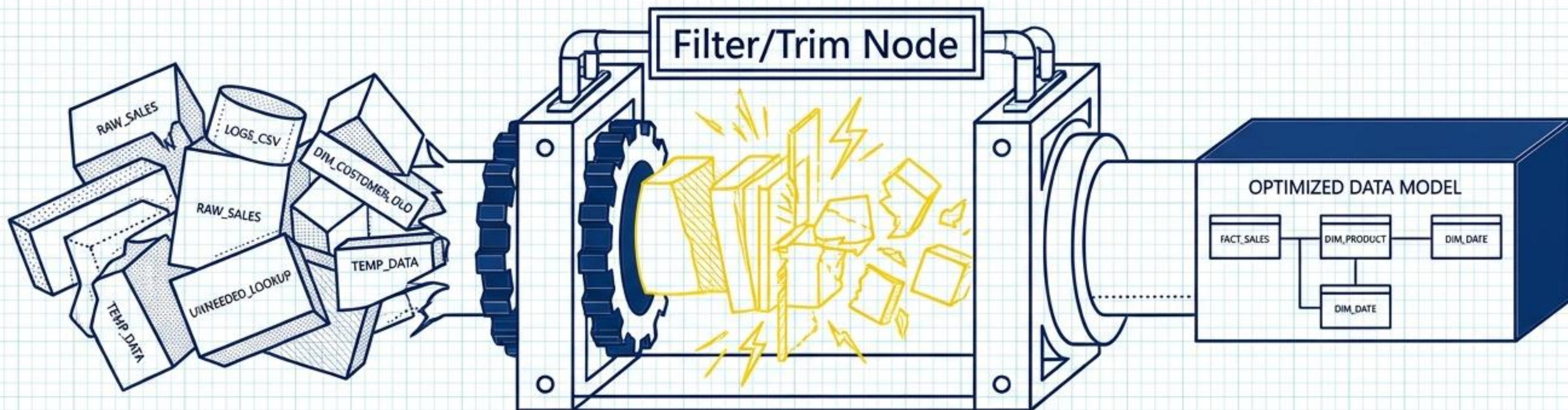
Power BI: Advanced ETL

Architecting robust, future-proof
data transformations in Power Query

dr Zbigniew Galar



The ETL Philosophy: Be Ruthless



The Power Query Editor is a forgiving environment, but your data model should be ruthless.

Strip the Excess:

Remove unused columns immediately.

Consolidate:

Combine partial queries via append or merge.

Justify the Load:

Never bring a query or column into the Desktop just in case. You can always come back for it later.

Augmenting Data: Generating New Columns

Path A: Column from Examples



- **Function:** Type the desired value; Power BI determines the pattern and suggests the rest.
- **Best for:** Rapid, intuitive string extraction and pattern matching.

Path B: Custom Column



- **Function:** Manually enter a formula written in M script.
- **Best for:** Precise control, complex conditional logic, or adding literal text strings.

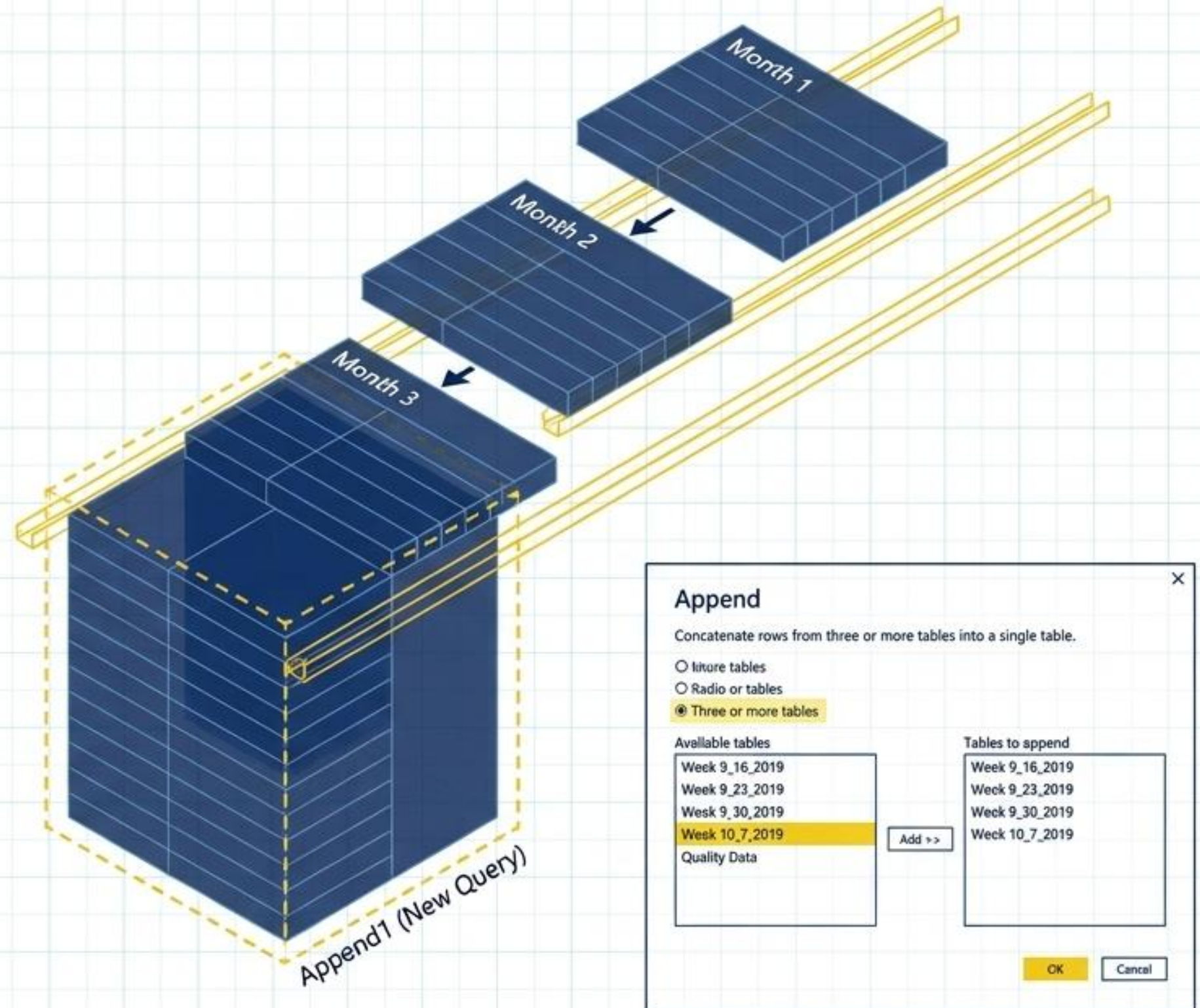
Combining Data Vertically: Appending Queries

Core Concept

Appending concatenates rows. It is ideal for stacking identically structured files (e.g., rolling monthly reports) into a single master dataset.

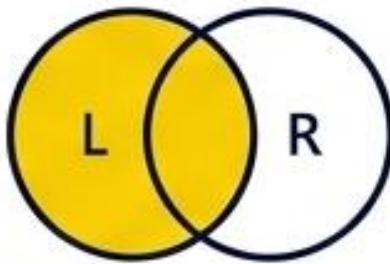
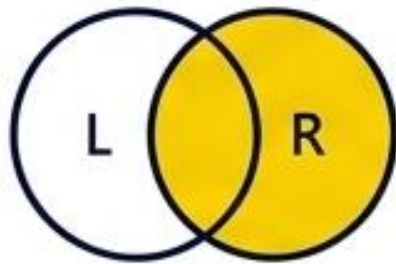
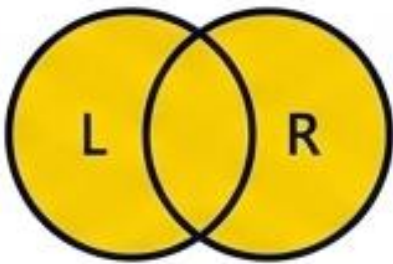
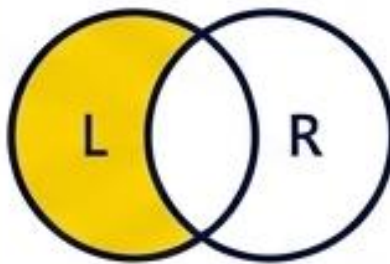
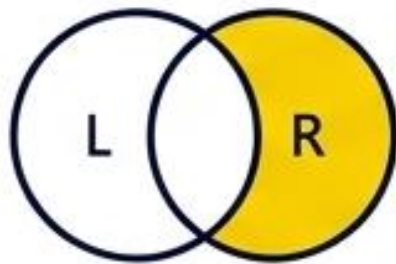
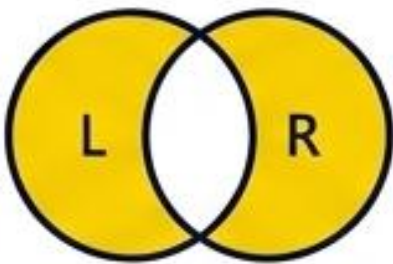
Architectural Best Practice

Always append into a New query (e.g., Append1). This preserves the original source queries in their unadulterated state for easier auditing and troubleshooting.



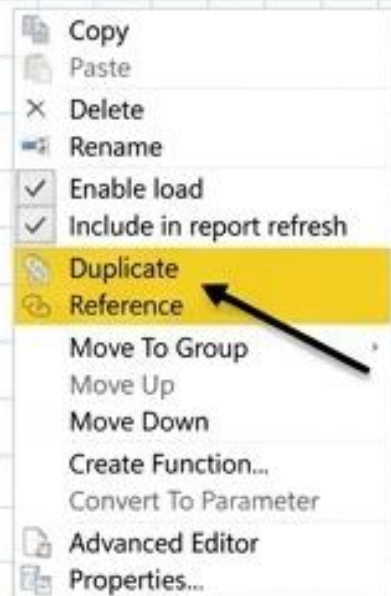
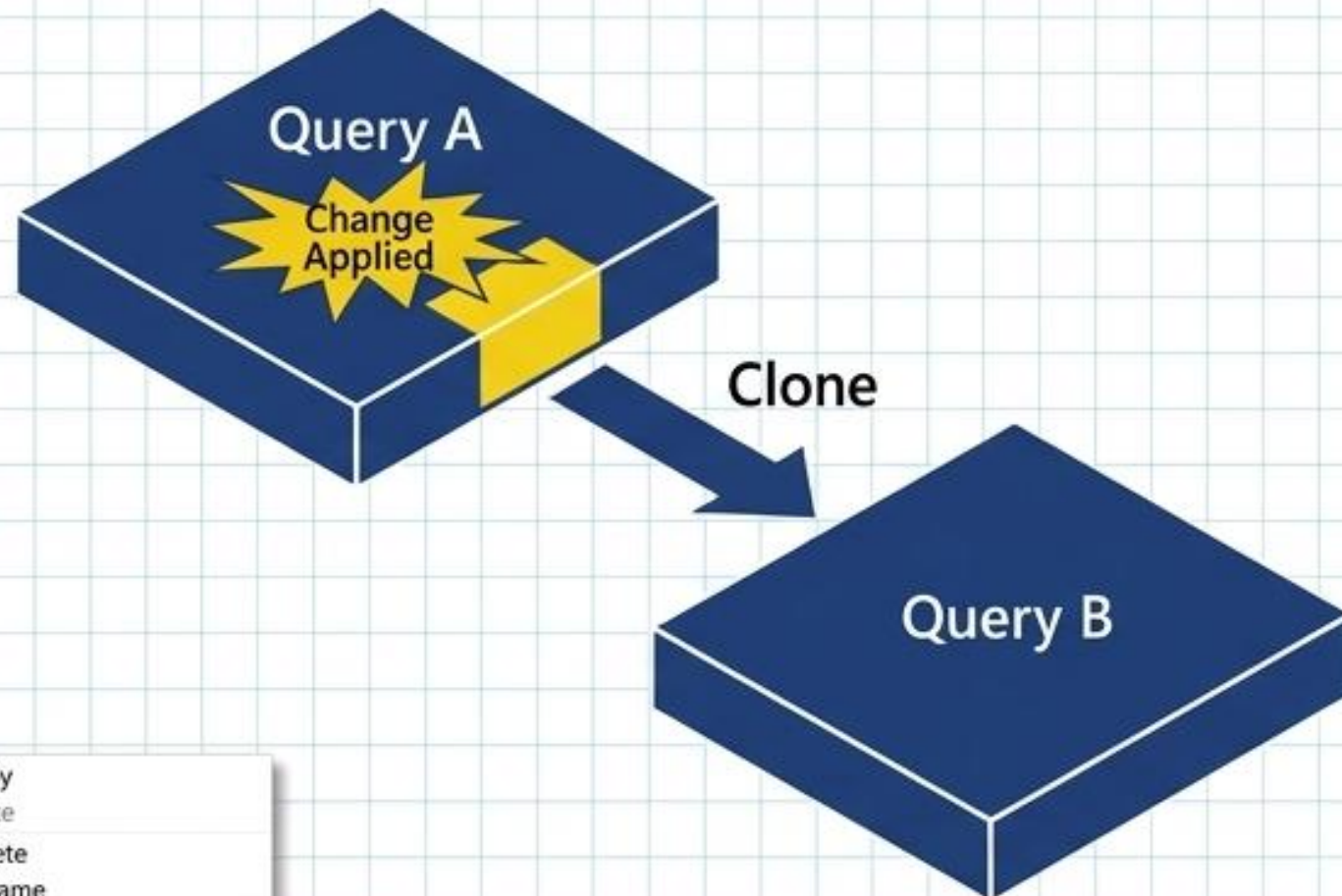
Combining Data Horizontally: The Join Matrix

Merging expands columns based on matching IDs. Select the correct join pattern to act as a gatekeeper for incoming rows.

	Left	Right	Full																																													
Outer	 All left, matching right Query L <table><tr><td>1</td><td>A</td></tr><tr><td>2</td><td>B</td></tr></table> Query R <table><tr><td>1</td><td>X</td></tr><tr><td>3</td><td>Y</td></tr></table> Merged Query <table><tr><td>1</td><td>A</td><td>X</td></tr><tr><td>2</td><td>B</td><td>null</td></tr></table>	1	A	2	B	1	X	3	Y	1	A	X	2	B	null	 All right, matching left Query L <table><tr><td>1</td><td>A</td></tr><tr><td>2</td><td>B</td></tr></table> Query R <table><tr><td>1</td><td>X</td></tr><tr><td>3</td><td>Y</td></tr></table> Merged Query <table><tr><td>1</td><td>A</td><td>X</td></tr><tr><td>3</td><td>null</td><td>Y</td></tr></table>	1	A	2	B	1	X	3	Y	1	A	X	3	null	Y	 All rows from both Query L <table><tr><td>1</td><td>A</td></tr><tr><td>2</td><td>B</td></tr></table> Query R <table><tr><td>1</td><td>X</td></tr><tr><td>3</td><td>Y</td></tr></table> Merged Query <table><tr><td>1</td><td>A</td><td>X</td></tr><tr><td>2</td><td>B</td><td>null</td></tr><tr><td>3</td><td>null</td><td>Y</td></tr></table>	1	A	2	B	1	X	3	Y	1	A	X	2	B	null	3	null	Y
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Query Lineage: Duplicate vs. Reference

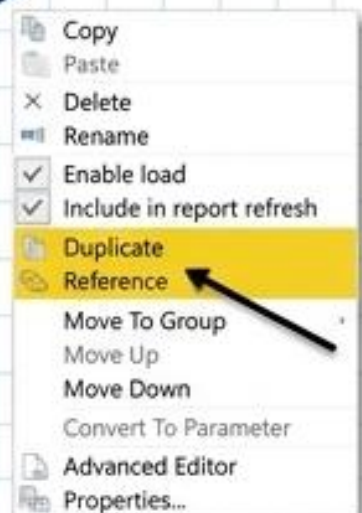
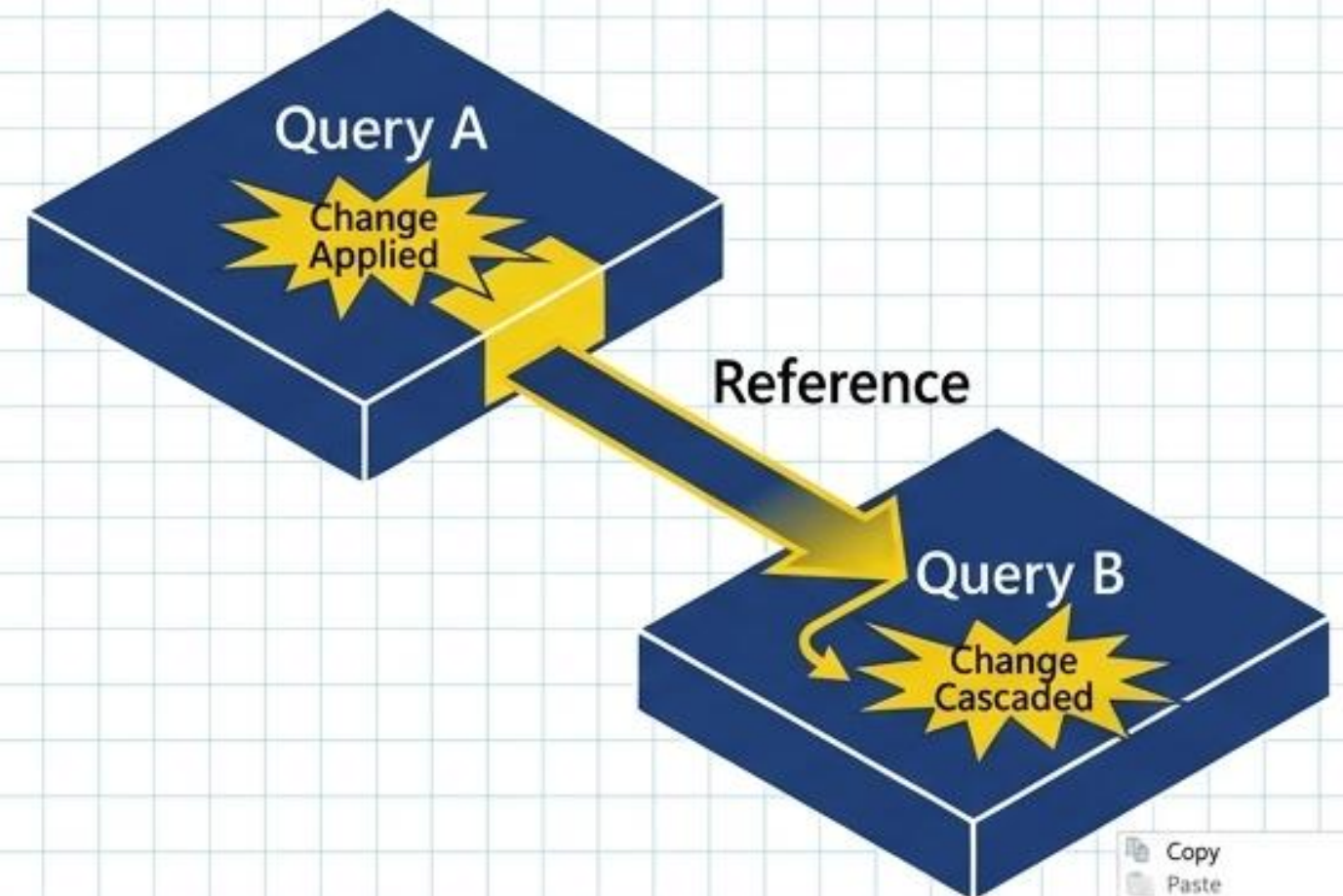
Left Track: Duplicate (The Isolated Path)



Definition: Creates an exact, independent copy.

Use Case: When you need to preserve the original query exactly as it is and diverge completely.

Right Track: Reference (The Cascading Path)



Definition: Creates a linked query. Upstream transformations automatically cascade down.

Use Case: When building multi-stage logic from a single, continuously updated base query.

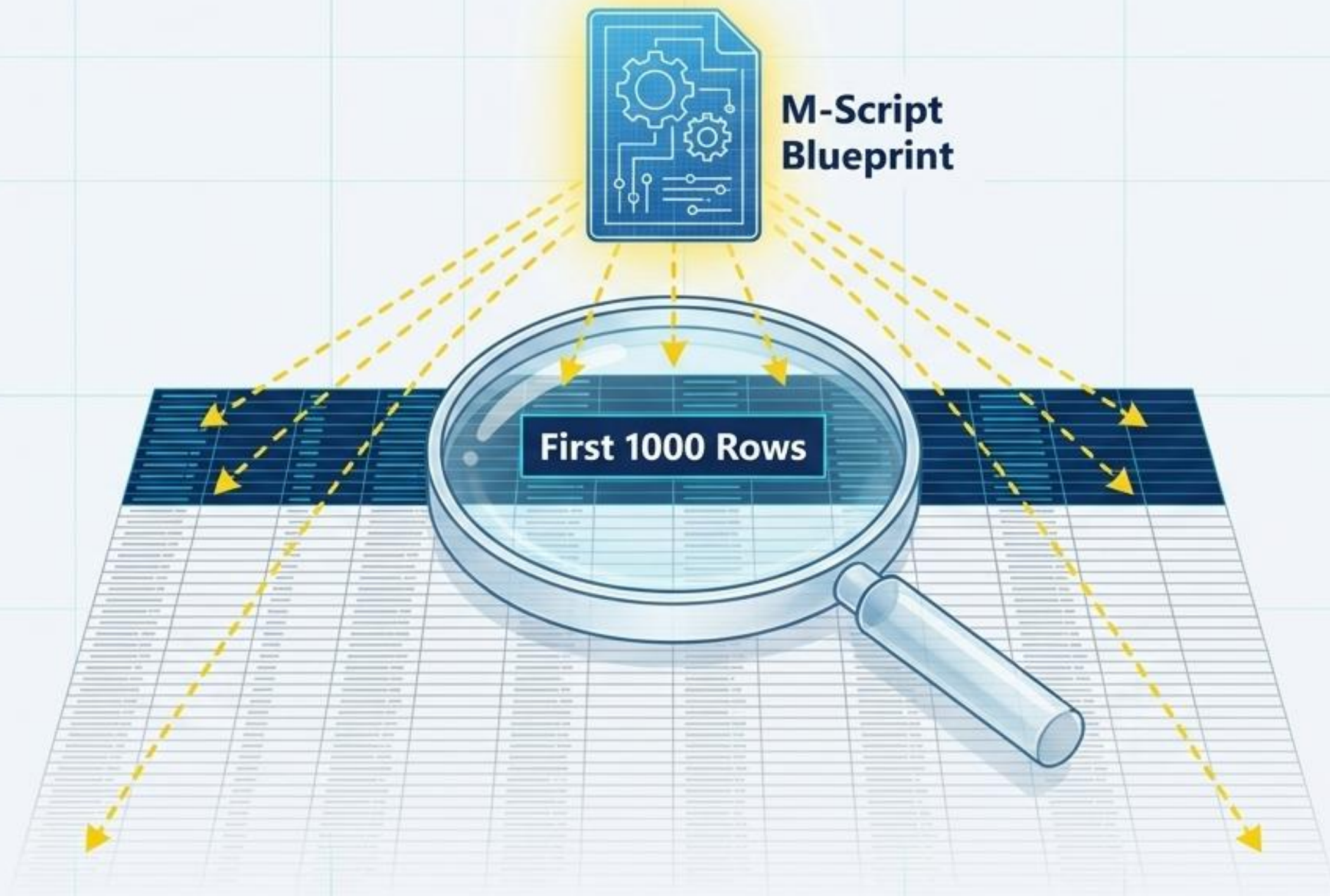
Navigating the 1000-Row Illusion

The Problem

The PQE interface only scans and previews the first **1,000** rows, leading to fears of incomplete data processing.

The Reality

PQE is not transforming data live. It is writing a script of transformations based on a representative sample. When loaded into Power BI Desktop, this exact script will execute against all millions of rows flawlessly.



Overcoming Filter Limits

The Constraint:

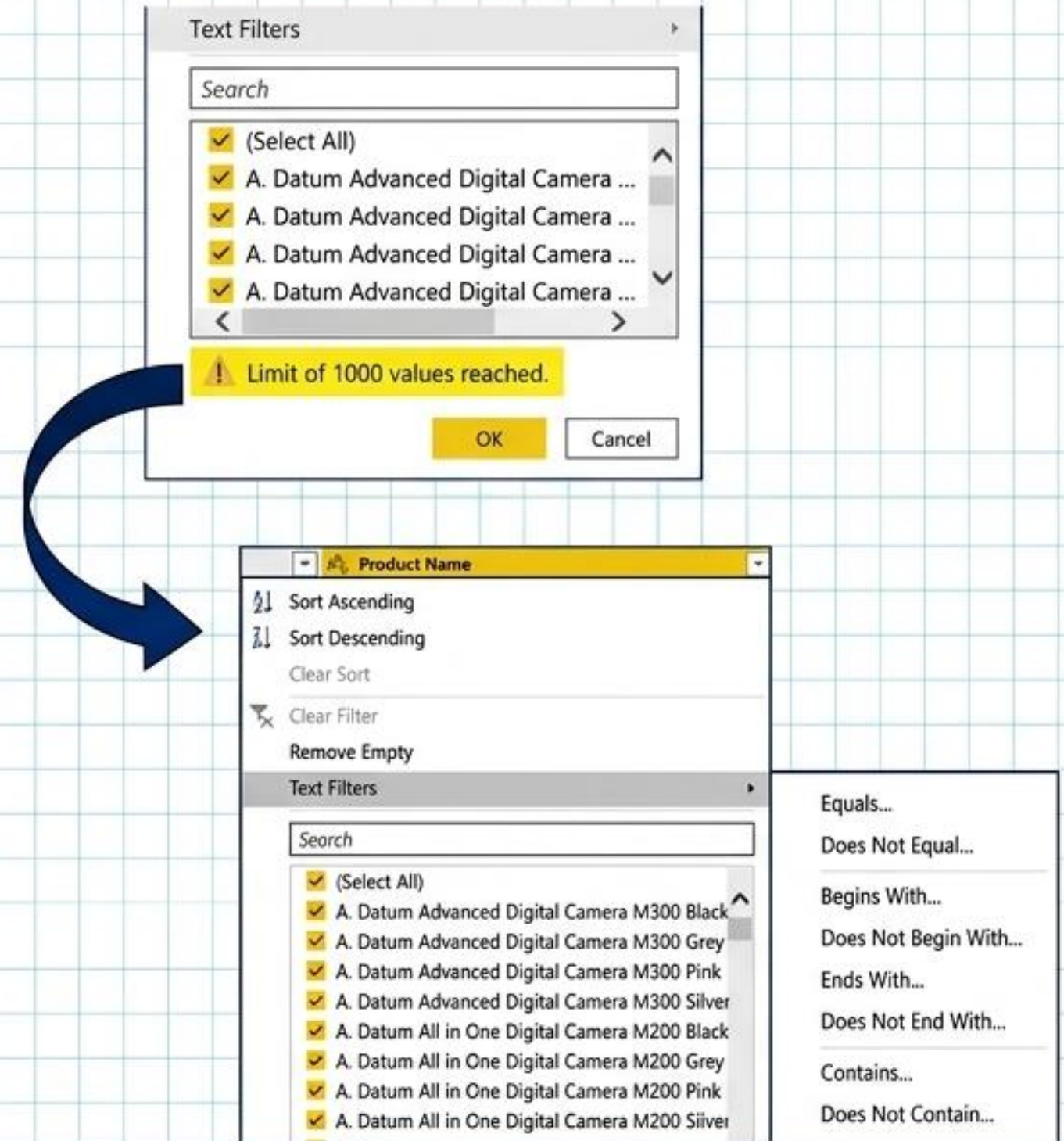
The standard Filters panel cannot display more than 1000 unique values for checkbox selection.

The Tactical Fix:

Abandon checkboxes. Switch to explicit **Manual Text/Numeric Filters** (e.g., Equals..., Begins With..., Contains...).

Pro-Tip:

Rename and explicitly document any step where rows are purposefully filtered out, as they will not appear in the final report.

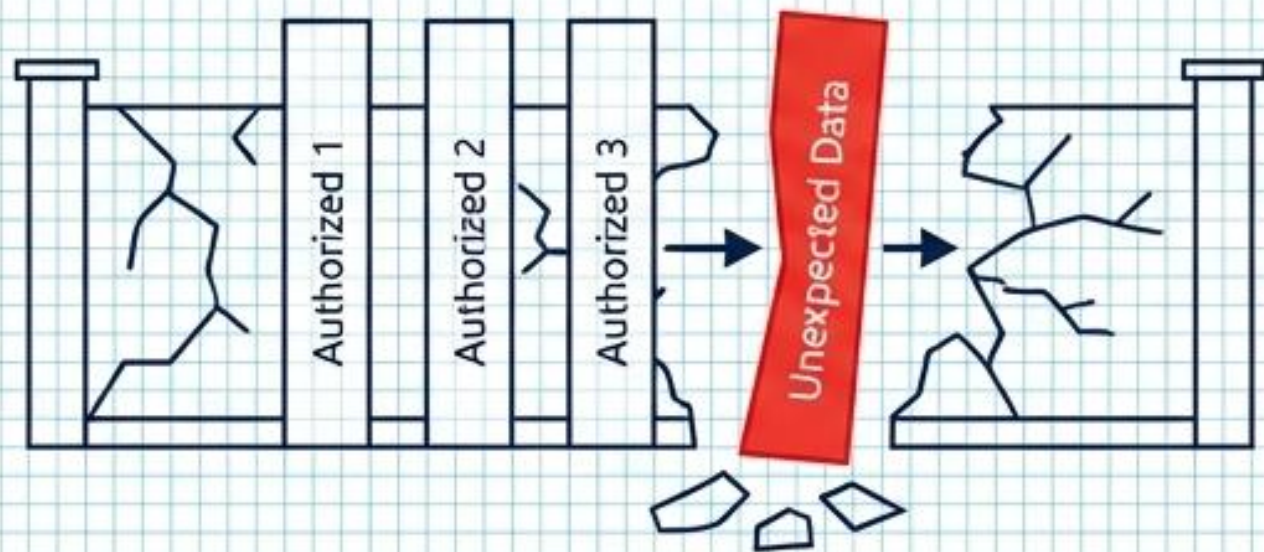


Future-Proofing: Fragile vs. Robust M-Code

The Assumption: Source files will mutate. Columns will be added, renamed, or deleted. Keep the Formula Bar open and read the generated M script.

Fragile Approach (Exclusionary)

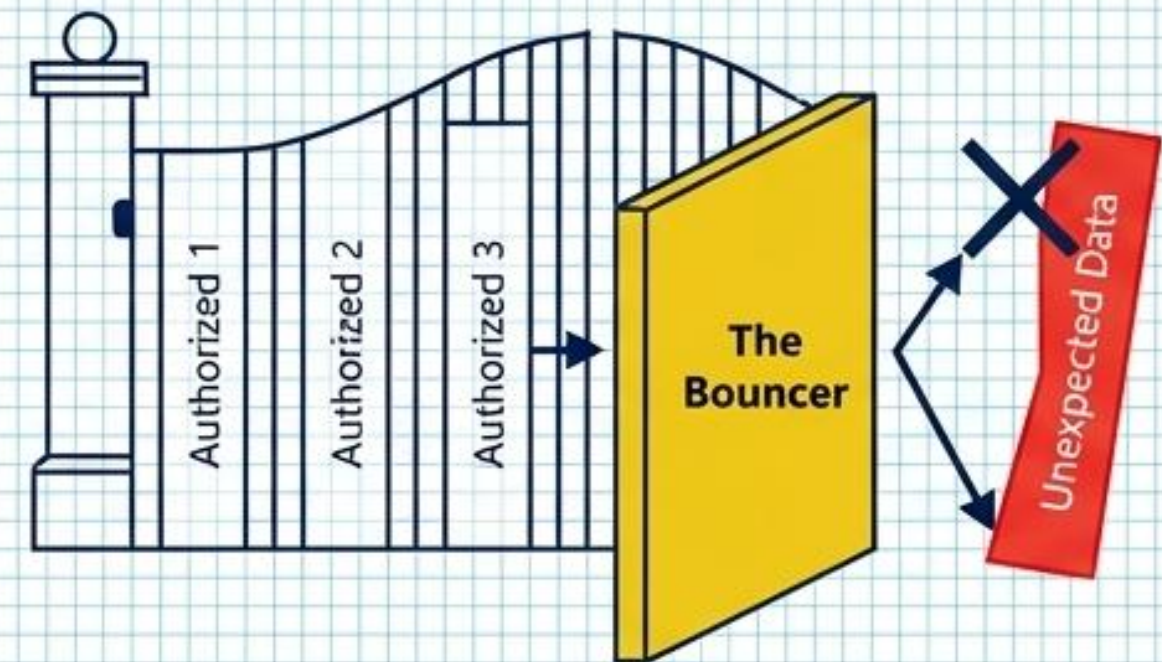
```
= Table.RemoveColumns(Source,  
  {"Vendor Number"})
```



Risk: If a user adds a rogue new column to the source Excel file, it slips straight through to the final report.

Robust Approach (Inclusionary)

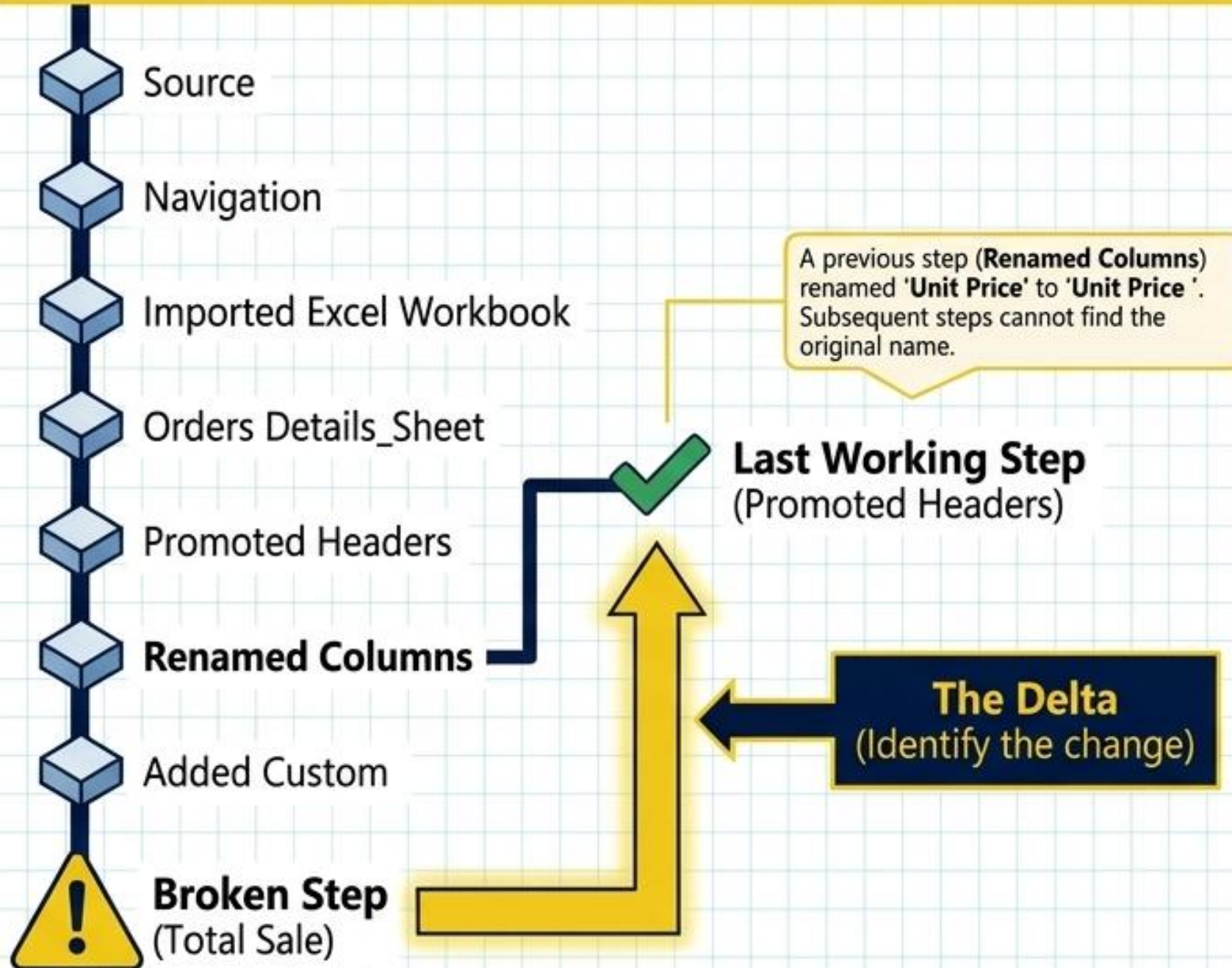
```
= Table.SelectColumns(Source,  
  {"Vendor Name", "CommentText.1"})
```



Reward: Acts as a strict bouncer. Only explicitly named columns pass through; surprise additions are automatically blocked.

Troubleshooting: The Error Resolution Framework

 Expression.Error: The field 'UnitPrice' of the record wasn't found.



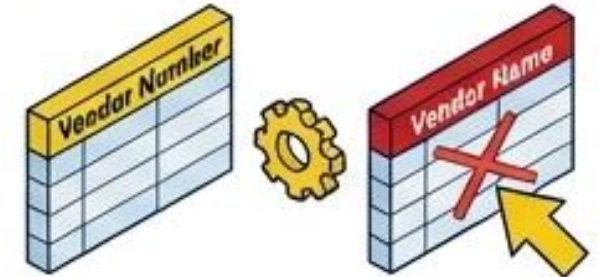
The Diagnostic Workflow

1. Start at the step displaying the yellow error bar.
2. Move backward (upward) step-by-step.
3. Stop at the last functional step.
4. Identify the delta (the exact change that broke the sequence).

The Two Usual Suspects

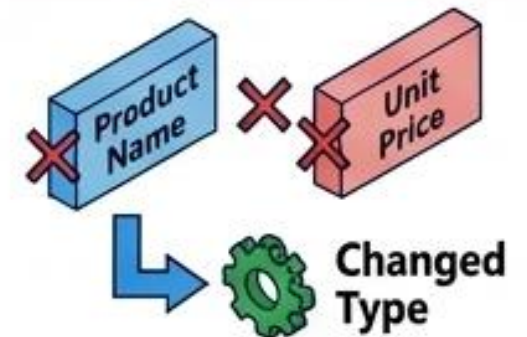
Missing/Renamed Column:

A previous step renamed a column that a subsequent step requires.



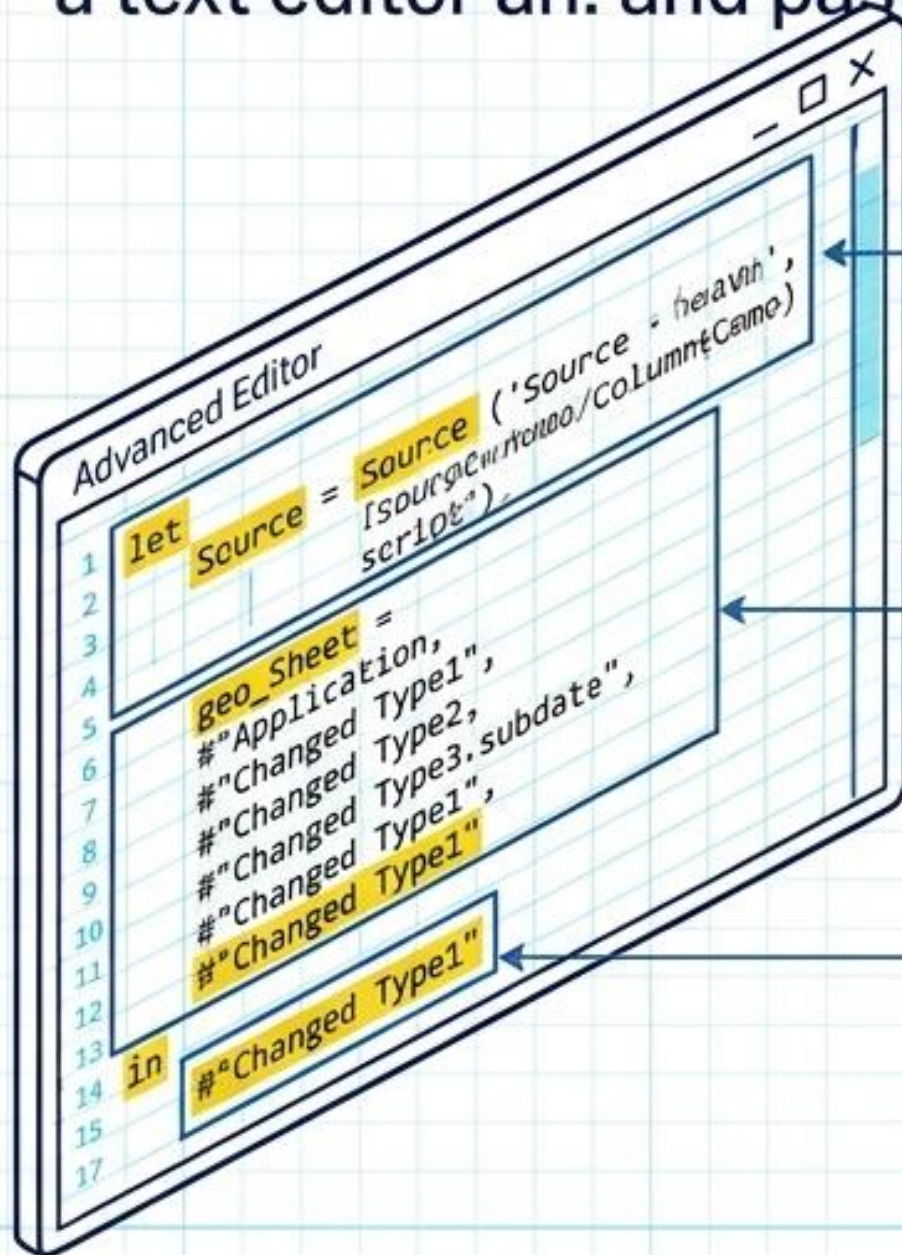
Incorrect Data Type:

An operation was attempted on incompatible types (e.g., multiplying text). Fix by inserting a **Changed Type** step immediately prior.



Reusability: The Advanced Editor

A well-written M script is an asset. Entire scripts or partial steps can be copied via a text editor and pasted into new files.



The Engine
(Variables & Setup)

The Gears
(Transformations)

The Exhaust
(Output)



The Migration Checklist
(Gotchas)



Syntax Check: Ensure a comma trails every step except the final one before 'in'.



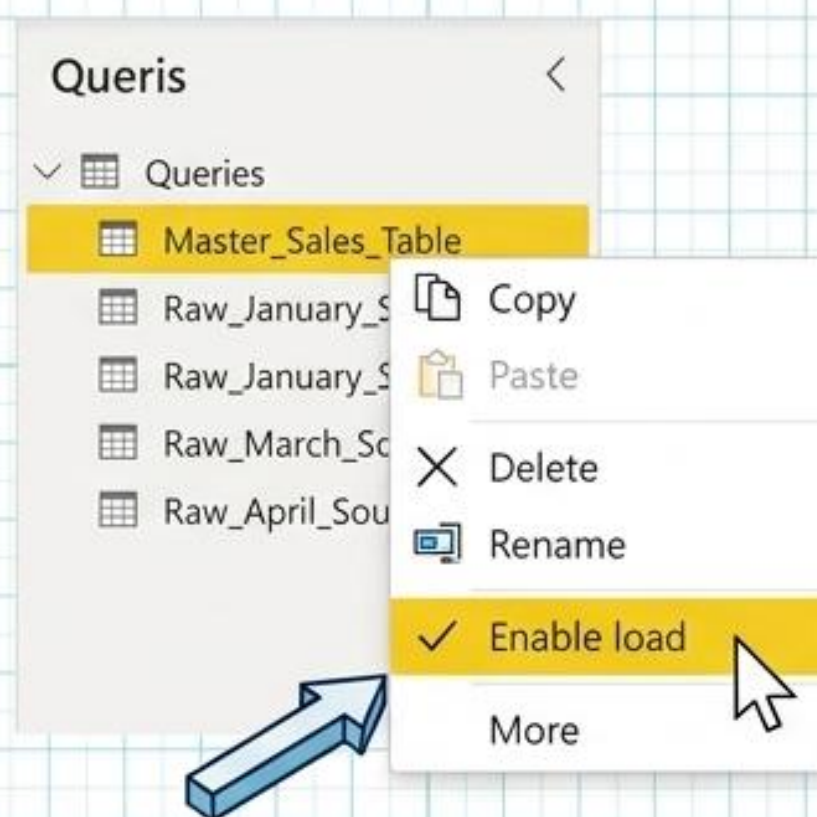
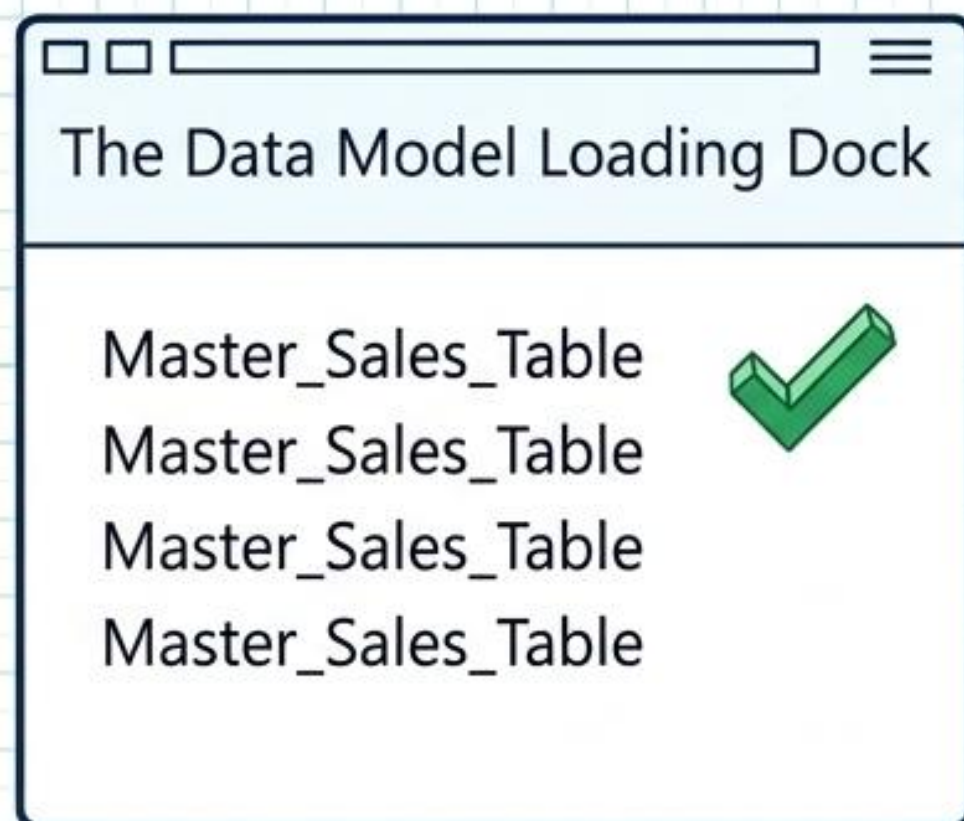
Nomenclature: Verify step names and column names match the new destination exactly.



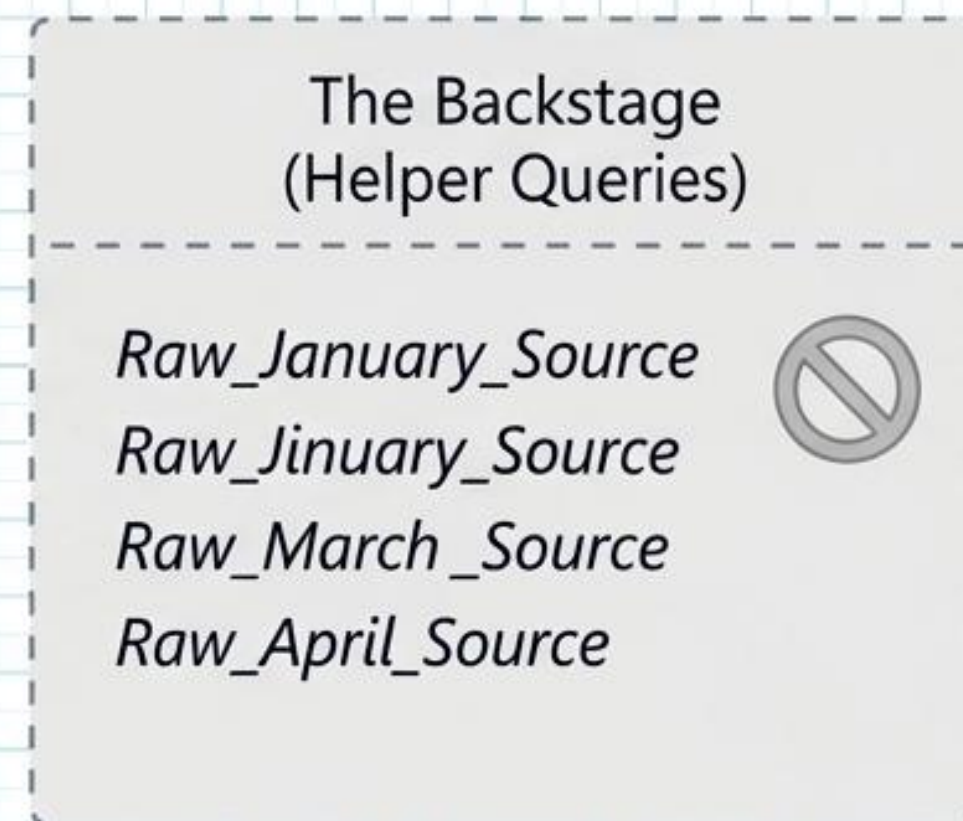
Source Verification: Ensure the first step ('Source') points to the correct new hard-coded file path.

Deployment: Managing the Load

Not every query needs to enter the Power BI Desktop data model.



The Action: Right-click the query and uncheck **Enable Load**. The query name will italicize. The logic remains intact, but the data model stays lightweight.



Queries used solely as intermediary steps should be **disabled** to optimize performance.

Summary: The 8 Golden Rules of Advanced ETL

